CSE320 (Data Comm) – Practice Sheet on Chapter 6

Multiplexing

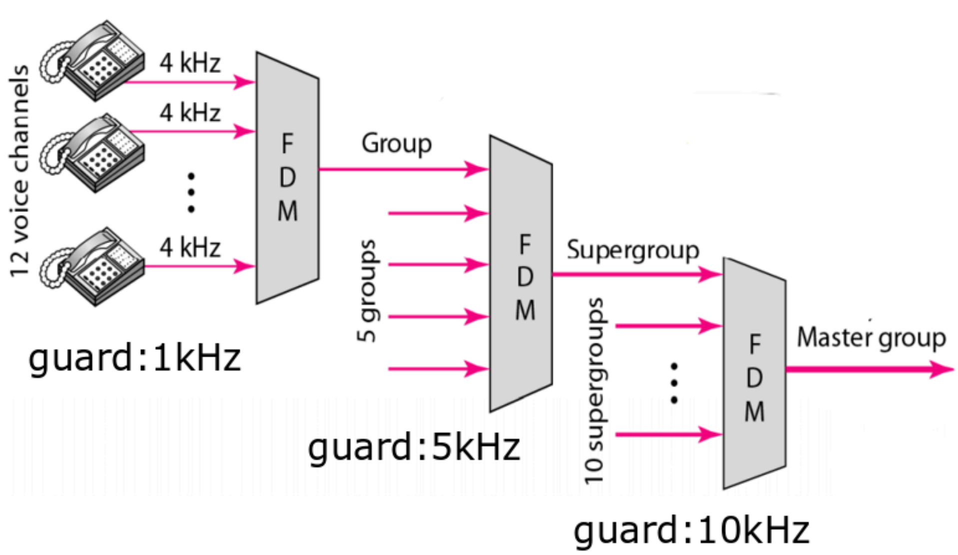
Q1.

Consider, Five channels, two with a bit rate of 240 kbps and three with a bit rate of 180 kbps, are to be multiplexed with one synchronization bit. Write the following answers:

I. What is the size of a frame in bits?
II. What is the frame rate?
III. What is the duration of a frame?
IV. What is the output data rate?
V. What is the output bit duration?
VI. How many input channels are there after doing multiplexing?

Q2.

The following FDM hierarchy has been used by a telephone company. How many voice channels can be multiplexed together in the master group? What is the required bandwidth for the multiplexing?



Q3.

Suppose you have two channels among which 1 channel has a bandwidth of 1500 kbps and one with 1200 kbps. What is the smartest way to multiplex these channels without involving too many extra bits? Draw and validate with visual representation to aid your reasoning.

Q4.

Why is the guard band necessary to use in FDM and not in TDM? Assume twelve 5.2 kHz channels are multiplexed in a 69 kHz channel using FDM. Calculate the bandwidth of the guard bands. Illustrate with visual representation.

Q5.

Consider, you have 4 sources, each creating 250 characters per second. If the interleaved unit is two characters long and 1 synchronous bit is added to each frame, find the followings:

i. What is the size of a frame (in bits)?

ii. What is the frame rate (fps)?

iii. What is the input bit rate (in bps)?

iv. What is the output data rate (in bps)?

v. What is the output bit duration (in sec)?

Q6.

Suppose you have three channels among which two channels have a bandwidth of 1600 kbps and one with 1200 kbps. How would you multiplex this without using any extra pulses? Draw and validate with visual representation.

Q7.

Suppose 5 data sources with data rates of 240 Kbps, 120 Kbps, 120 Kbps, 240 Kbps and 480 Kbps are to be multiplexed with one synchronization bit. Show the visual representation of MUX with arbitrary input and output bits without adding extra pulses in the input channel.

Q8.

Suppose, each of the 5 channels send 6000 pages per minute where each page consists of 300 characters. If synchronous TDM is used to combine these sources using two-characters interleaving with one sync bit then answer the following questions:
i. What is the input data rate (in bps) for each of the connections?
ii. What is the input bit duration (in seconds)?
iii. What is the frame rate (frames per second)?
iv. What is the size of a frame in bits?
v. What is the output data rate (in bps)?
vi. What is the output bit duration (in seconds)?

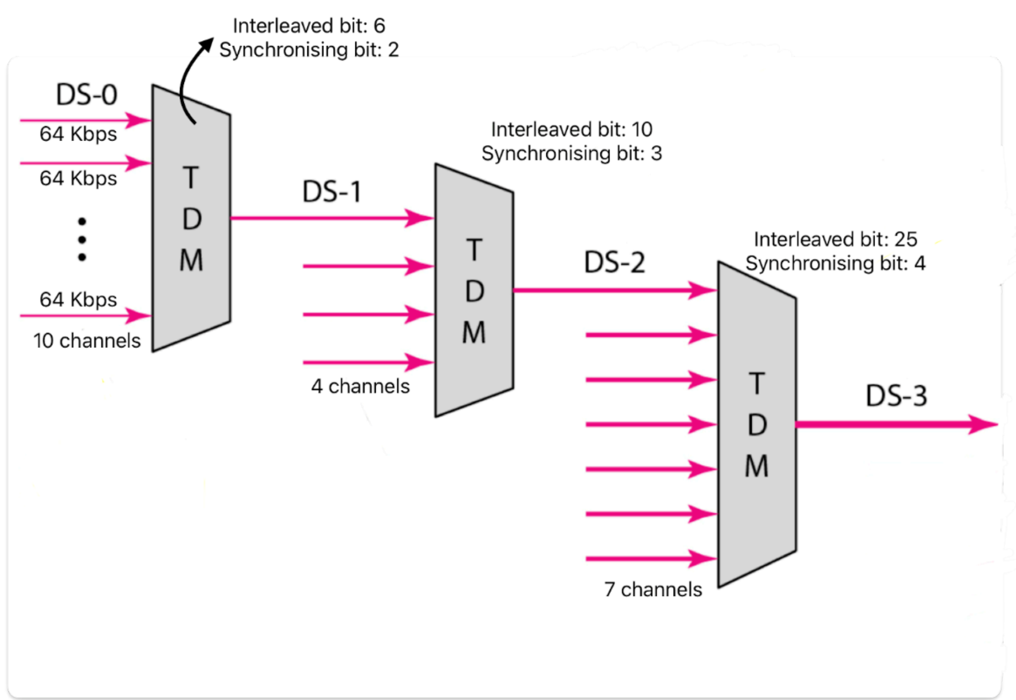
Q9.

A control system has 8 devices. In the first time slot, only devices 1, 3, 5, 6, and 8 have data. In the second time slot, devices 1, 2, 4, 5, and 7 have data.

● What TDM technique should be used here?
● Sketch a figure of the TDM process and show the frames.
● If three bits are used for addressing, determine the frame size.
● Does this scheme have any drawbacks? Justify.

Q10.

In this hierarchical TDM, calculate the frame rate and output rate for DS-3.



Q11.

In a FDM, 4 input channels are multiplexed together. The size of the assigned bandwidth for each channel are 5KHz, 10 KHz, 15 KHz and 20 KHz respectively. If the guard band is 3KHz. What is the overall required bandwidth of the output channel?

Q12.

The minimum bandwidth required to multiplex 6 channels is 555 kHz. There are 15 kHz guard bands used to prevent interference for each channel. Calculate the bandwidth of each input channel. Show the required bandwidth with a diagram where the starting frequency is 101 kHz. What will be the highest frequency?

Q13.

Consider 3 channels, each having different input data rates. If the data rate of the first channel is 12 kbps, then the rest of the channels have double the data rate of their previous channel. You need to multiplex them in such a way, so that the number of channels increases. The interleaved unit is 4 bits along with 1 bit for synchronization. Calculate the following:

i. Draw the figure to visualize the multiplexing process with proper labels
ii. Input bit duration
iii. Frame Size
iv. Frame Rate
v. Output bit duration

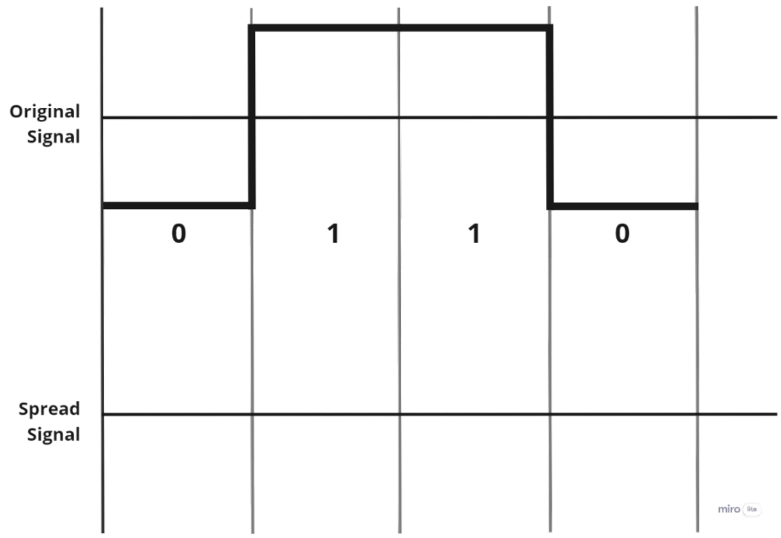
Spectrum Spreading

Q14.

What is the minimum number of bits in a PN sequence if we use FHSS with a channel bandwidth of B = 5Hz and bandwidth of spread spectrum Bss = 250 KHz.

Q15.

Assume we want to send the following signal using DSSS technique. For ensuring higher security, we have invented a 5-bit spreading code “10110”. Draw the corresponding spread signal on the question paper. You can use bipolar NRZ encoding (0 = negative voltage, 1 = positive voltage) for signal drawing. **Comment** on the bandwidth of the spread signal in brief.



Q16.

Suppose, you are given with the k-bit pattern and Carrier Frequency as follows:

k-bit pattern

11 00 01 10

k-bit	Carrier Frequency
00	100 kHz
01	300 kHz
10	400 kHz
11	200 kHz

Draw FHSS cycle 2 times using the above pseudo random generated k-bit pattern and given frequency table. (** Hint: Draw the Carrier frequency graph against hop period)

Q17.

How does DSSS achieve bandwidth spreading and provide privacy? Sketch the Spread Signal from the following Original Signal and the given spreading code.